SkillProof Engineering Team

Technical Roles, Ownership & Deliverable Contributions Breakdown

Platform Scope: SkillProof is a deterministic, quality-gated problem-solving platform tailored for the education sector (coaching hubs, tuition academies, and college departments). The system connects student solvers with institutions by replacing unstructured freelance proposals with a standardized 5-step pitch wizard, a two-tier grading gate (binary knockouts + 6 quality dimensions at a 6.0/10.0 threshold), and noise-free vetted inboxes with 1-click PDF export. Below is the comprehensive responsibility and contribution profile for each of the 5 core team members.

Nitin — Frontend Engineer

Primary Ownership: User Interface, Stepper Navigation, Interactive Gauges & Client State

Key Code / Artifacts: `components/Stepper.tsx, ScoreMeter.tsx, KnockoutCard.tsx, ProposalCard.tsx, SlideButton.tsx, app/problems/[id]/pitch/page.tsx`

Key Technical Deliverables & Contributions:

- **5-Step Guided Pitch Wizard:** Built the comprehensive solver workflow breaking proposals into 5 structured stages (Problem Breakdown, Technical Architecture, Alternative Analysis, Budget/Milestones, and Proof/Feasibility).
- **Vertical Stepper (Uiverse Component):** Developed the dynamic vertical stepper component tracking real-time status badges (Completed, In Progress, Pending) and connecting lines.
- **6-Tube Evaluation Gauge:** Created the cylindrical 6-dimension score meter displaying visual fluid fill heights and dynamic threshold colors (Emerald for ≥ 6.0 pass, Rose for < 6.0 revision).
- **Comic Neo-Brutalist Knockout Card:** Implemented the high-impact diagnostic card rendering when Tier 1 fails, displaying exact gate violations and direct 'Fix Step X' revision CTAs.
- **Vetted Proposal Cards & Poster Inbox:** Structured the client-side proposal cards featuring verified solver badges, fixed price tags, delivery turnaround, and 1-click standardized PDF downloads.
- **Interactive Slide-Fill Buttons:** Standardized the smooth slide-fill button animations across all hero banners, wizard controls, auth pages, and inbox action bars.

Tanmay — Backend Engineer

Primary Ownership: Deterministic Scoring Algorithm, Database Relational Schema & Core APIs

Key Code / Artifacts: `lib/scoring/engine.ts, prisma/schema.prisma, lib/services/pitch.service.ts, problem.service.ts, app/api/*`

Key Technical Deliverables & Contributions:

- **Two-Tier Quality Gating Engine:** Engineered the pure, stateless, zero-external-LLM grading algorithm executing in under 10ms with mathematical determinism.
- **Tier 1 Binary Knockout Logic:** Built zero-tolerance knockouts immediately disqualifying submissions (Score = 0) if pricing is missing, fewer than 2 alternatives are evaluated, or total length is under 200 characters.
- **Tier 2 6-Dimension Quality Scorer:** Programmed the weighted 10-point scoring algorithm covering Problem Diagnosis (2.0), Solution Feasibility (2.0), Alternative Depth (1.5), Pricing Transparency (1.5), Solver Credentials (1.5), and Constraints/Edge Cases (1.5).
- **Relational Multi-Tenant Data Schema:** Architected the Prisma SQLite/PostgreSQL schema modeling Users, Organizations, Problems, and Pitches scoped by organizationId.
- **API Services & Endpoint Architecture:** Developed decoupled backend services and Next.js Route Handlers powering pitch submissions, problem creation, and automated credibility score updates.

Sameer — Testing & QA Lead

Primary Ownership: Quality Assurance, Calibration Benchmark Runner, Edge Cases & E2E Validation

Key Code / Artifacts: `lib/scoring/calibrationData.ts, scripts/calibrate.ts, app/admin/calibrate/page.tsx, E2E Test Suite`

Key Technical Deliverables & Contributions:

- **Calibration Benchmark Suite (VAL-1 to VAL-6):** Authored the definitive test corpus spanning high-quality passes (≥ 7.0), knockout edge cases (missing quotes, 1 alternative, low effort), borderline fails (< 6.0), and minimal passing thresholds (6.0-6.5).
- **Automated Calibration CLI Runner:** Built the standalone calibration test runner (scripts/calibrate.ts) allowing instant automated regression testing of scoring accuracy.
- **Admin Gate Verification Dashboard:** Tested and verified the interactive /admin/calibrate portal enabling evaluators to run real-time gate benchmarks with zero score drift.
- **End-to-End User Flow Validation:** Executed thorough validation of both core loops: Solver Journey (Discovery → 5-step draft → Gate scoring → Diagnostic feedback → Revision loop) and Poster Journey (Problem post → Vetted inbox → Noise shielding audit → PDF export).
- **Type Safety & Build Integrity:** Enforced strict zero-error TypeScript compilation across all app routes and component interfaces (npx tsc --noEmit).

Nikhil — Security Engineer

Primary Ownership: Multi-Tenant Data Isolation, Input Validation, Anti-Gaming & Boundary Defense

Key Code / Artifacts: `lib/api/response.ts, lib/api/errors.ts, lib/auth/AuthContext.ts, API Middleware`

Key Technical Deliverables & Contributions:

- **Multi-Tenant Data Isolation:** Enforced query-level organizational boundary constraints ensuring problems, pitches, and candidate proposals remain strictly segregated by organizationId.
- **Input Sanitization & Schema Defense:** Built strict input validation and centralized error handling (AppError, withErrorHandler) to prevent payload tampering, script injections, and stack trace leaks.
- **Anti-Gaming Deterministic Gate Rules:** Hardened the scoring engine against buzzword stuffing by verifying structural integrity in trade-off comparisons (distinct pros, cons, and rejection reasons).
- **Pricing Quote Integrity:** Blocked solver attempts to bypass budget knockouts with placeholder text ('negotiable', 'discuss later'), enforcing numeric or currency-explicit quotes.
- **Local Authentication & IAM Roadmap:** Implemented role-based authorization (SOLVER, POSTER, ADMIN) with clean passwordless local session handling, primed for seamless Google OAuth migration.

Tushar — Hosting & DevOps Engineer

Primary Ownership: Google Cloud Infrastructure, Containerization, Domain/SSL & Production Ops

Key Code / Artifacts: `Dockerfile, next.config.mjs, GCP Cloud Run / Compute Engine Config, Deployment Scripts`

Key Technical Deliverables & Contributions:

- **Google Cloud Server Deployment:** Configured and deployed the production Next.js application on Google Cloud infrastructure (Google Cloud Run / Compute Engine / Firebase App Hosting).
- **Environment & Secrets Management:** Managed server environment variables (DATABASE_URL, NODE_ENV, port mappings) and secure runtime configurations across cloud environments.
- **Build Artifact Optimization:** Optimized Docker container builds and Next.js standalone outputs, minimizing image size and eliminating extraneous dependencies.
- **Domain, DNS & SSL Provisioning:** Managed custom domain routing, DNS record setups (A / CNAME records), and automated Google-managed SSL/TLS certificates for HTTPS encryption.
- **Uptime & Latency Monitoring:** Configured cloud health probes, cold-start mitigation, and container resource scaling to guarantee responsive performance during evaluator demonstrations.

Consolidated Team Contribution Matrix

Quick reference matrix mapping team members to their core deliverable and technical stack.

Member	Role	Core Deliverable	Tech Stack & Key Files
Nitin	Frontend Engineer	5-Step Pitch Wizard, 6-Tube Gauge, Stepper, Knockout Card	React, Next.js, Tailwind, components/
Tanmay	Backend Engineer	Two-Tier Deterministic Scoring Engine, Relational DB, API Handlers	TypeScript, Prisma, SQLite, lib/scoring/
Sameer	Testing & QA Lead	VAL-1 to VAL-6 Calibration Test Harness, Admin Portal, E2E QA	Benchmark Suite, scripts/calibrate.ts, QA
Nikhil	Security Engineer	Anti-Gaming Gate Guards, Input Sanitization, Multi-Tenancy	Schema Validation, Security Rules, Auth

Tushar	Hosting & DevOps	Google Cloud Server Deployment, Docker, SSL, DNS, Performance	GCP Cloud Run / Compute, Docker, CI/CD